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Professional Summary

Python and agentic AI engineer with **3+ years** shipping **FastAPI** services, **LangGraph** and **CrewAI** multi-agent orchestration, **n8n** workflow automation, and retrieval-augmented generation (**RAG**) on **Azure OpenAI**, **Azure AI Search**, and **GCP Cloud Run**, with human-in-the-loop guardrails and LLM evaluation that cut hallucinations **14% to 3%**.

Skills

Backend & Data: Python (asyncio, Pydantic), **FastAPI**, Flask, SQLAlchemy, **PostgreSQL**, Redis, Celery, Kafka, Spark
Agentic AI: LangGraph, **CrewAI**, **n8n**, Multi-Agent Orchestration, Tool Calling, **MCP** Servers, Human-in-the-Loop
RAG & LLMs: RAG, Hybrid Search, Vector Embeddings, Reranking, NL2SQL, **Azure OpenAI**, **Ollama**, **vLLM**
LLMOps: LLM Evaluation (Ragas), Langfuse, Guardrails, PII Redaction, PyTorch, PEFT, **QLoRA**, **AWQ**, GPTQ, GGUF
Cloud: GCP (Cloud Run, Cloud Functions, Pub/Sub), **Azure** (AKS, Key Vault), **Docker**, **Kubernetes**, **Terraform**

Work Experience

Graduate Research Assistant | **Stony Brook University** | Stony Brook, NY **Sep 2025 – Present**

- **Built** a **Python RAG** service with hybrid search (BM25 plus bge-base vector embeddings), **Neo4j** entity-graph retrieval, and cross-encoder reranking over **50K+** papers for **40+ researchers**, raising precision@5 from **0.61 to 0.86**.
- **Served** 4-bit **AWQ** Qwen2.5-7B on **vLLM**, cutting VRAM **15.2 to 5.6 GB** within **1** point; routing to it cut cost **58%**.
- **Scored** groundedness with **Ragas** and LLM-as-judge tests in GitHub Actions; guardrails cut hallucinations **14% to 3%**.
- **Designed** event-driven ingestion on **GCP**: Cloud Storage uploads publish to **Pub/Sub** and trigger **Cloud Functions**, with the **FastAPI** API on **Cloud Run** and model observability in Prometheus, falling back below **0.8** confidence.

AI Engineer Intern | **Lumina View** | New York, NY **May 2025 – Aug 2025**

- **Built CrewAI** multi-agent orchestration in **Python** (classifier, extractor, and validator agents) that extracts fields from **50K+ documents per month**, running Qwen2.5 on **Ollama** on an Azure T4 GPU VM with **Azure OpenAI** fallback.
- **Defined** agent tool calling with JSON-schema structured outputs and role-based tool permissions, redacting PII with Presidio and writing every tool call and approval to a **PostgreSQL** audit log for **100%** of runs.
- **Automated 3 client workflows** in self-hosted **n8n** on Docker Compose, with webhook triggers, REST API integrations into client systems, retry-on-fail, error workflows, and human-in-the-loop approval before writes.
- **Served** agents via async **FastAPI** and **Celery** on **Redis**, tracing in **Langfuse** to fix **20+** production failures.

Software Engineer | **Mavdero Techservices** | Chennai, India **Jun 2022 – Jun 2024**

- **Integrated 12** CRM and billing sources into **PostgreSQL** via **Python** ETL pipelines with JSON Schema validation.
- **Built Spark** Structured Streaming and **Kafka** pipelines in **Java** and **Python** for **20M+ events daily**.
- **Tuned PostgreSQL** for **3M+ daily requests** with EXPLAIN ANALYZE and indexes, cutting p95 latency **38%**.

Projects

Investor Intelligence Platform | *FastAPI, LangGraph, Azure OpenAI, Azure AI Search, AKS, Terraform*

- **Built** a **LangGraph** agent calling **MCP** server tools for NL2SQL on **PostgreSQL** and RAG over **500+** filings.
- **Indexed** filings via Document Intelligence into **Azure AI Search** hybrid indexes, scoring **0.90** Ragas faithfulness.
- **Deployed** to **AKS** with Helm, ACR, and Key Vault, provisioning **2** environments with **Terraform**.

Qwen Fine-Tuning and Quantization Benchmark | *PyTorch, PEFT, QLoRA, AWQ, GGUF, vLLM, llama.cpp*

- **Fine-tuned** Qwen2.5-1.5B with **QLoRA** for text classification, lifting macro-F1 **0.72 to 0.90**.
- **Quantized** it to **AWQ** and **GGUF** Q4_K_M (**3.1 to 1.0 GB**, F1 within **1** point), with **2.8x** faster CPU inference.

Education

Master of Science in Computer Science | **Stony Brook University** | Stony Brook, NY **Aug 2024 – May 2026**
Coursework: Distributed Systems, Machine Learning, Natural Language Processing, Database Systems